MATHEMATICS

Variance of X vs. X^2

Asked 6 years, 3 months ago Modified 3 years, 5 months ago Viewed 1k times

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> Let X be a discrete random variable and suppose $\mathrm{Var}(X) > 1$. How do $\mathrm{Var}(X)$ and $\mathrm{Var}(X^2)$ compare?"

Is this enough information to solve?

I think that $\mathrm{Var}(X^2)$ is larger than $\mathrm{Var}(X)$, but I only arrived at the answer by creating a hypothetical situation and solving it, but not sure if that holds in all cases or if there was a more obvious or intuitive way to come to this conclusion.

random-variables variance

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edited Dec 11, 2019 at 7:51
TheHolyJoker 2,060 1 9 25

asked Jun 28, 2018 at 16:53
Eliza Eliza Watts Sells 61 3

1 Answer

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2

If X takes values from $\{-1, 1\}$, then X has a positive variance but $X^2 = 1$ has zero variance.

On the other hand for a Bernoulli variable I, take $Y = aI$ then

$$\mathrm{E}(Y) = ap$$

$$\mathrm{E}(Y^2) = a^2p$$

$$\mathrm{E}(Y^4) = a^4p$$

So

$$\mathrm{Var}(Y) = a^2pq$$

$$\mathrm{Var}(Y^2) = a^4pq$$

So depending on a we will get any relation.

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edited Dec 10, 2019 at 12:47
Escalen 103 2

answered Jun 28, 2018 at 18:08
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